

Week 5 Lab: IP Security (IPsec)

Connect network-layer threat analysis with this week's topic: **IP Security (IPsec)**. You will review core IPsec concepts and apply them through hands-on observation (where available) and policy reasoning. All tasks should be completed inside **TryHackMe** (AttackBox or your own VM connected to THM).

Learning objectives

The aim of this lab is to help students understand why security at the IP layer remains important even when higher-layer protections already exist. By completing the activities, students will identify the main security services provided by IPsec, including confidentiality, integrity, authentication, access control, and anti-replay protection. They will also learn to distinguish between transport mode and tunnel mode and choose the most appropriate option for a given scenario. In addition, the lab introduces Security Associations (SAs) and explains the roles of the Security Policy Database (SPD) and Security Association Database (SAD). Finally, students will apply these concepts by creating simple IPsec-style policy rules for common enterprise use cases.

Requirements

- TryHackMe account and access to AttackBox or your own VM.
- Comfort with a terminal and basic packet inspection.

TryHackMe links

- Networking Secure Protocols: <https://tryhackme.com/room/networkingsecureprotocols>
- Network Security Protocols: <https://tryhackme.com/room/networksecurityprotocols>
- OpenVPN (THM connectivity guide): <https://tryhackme.com/room/openvpn>
- Intro to Security Architecture (context for network-layer security): <https://tryhackme.com/room/introtosecurityarchitecture>
- Networking Concepts (optional refresher): <https://tryhackme.com/room/networkingconcepts>